IT Troubleshooting Assistant

1. Project Description

The IT Troubleshooting Assistant is a Python-based application designed to help students solve common computer problems. It provides simple, step-by-step troubleshooting instructions for hardware, software, network, performance, and peripheral issues.

The system allows users to browse available problems, search using keywords, view recommended solutions, and provide feedback about whether the instructions were helpful.

2. Problem Statement

Students often experience computer problems but may not know how to identify the cause or solve them. Common issues include Wi-Fi connection problems, applications freezing, computers failing to start, slow performance, and printers not working.

Students may need to wait for technical support to resolve these problems, which can interrupt their studies and waste time.

3. Proposed Solution

The IT Troubleshooting Assistant provides a simple, menu-driven system that helps users find solutions to common computer problems independently.

Users can browse all available problems, select a category, or search using keywords. The system displays recommended troubleshooting steps and safety reminders. If a problem cannot be found or resolved, users are advised to try another search or contact IT support.

4. Main Features

- **Browse All Problems:** Displays all eight predefined troubleshooting problems.
- **Browse by Category:** Organises problems into Hardware, Network, Software, Performance, and Peripheral categories.
- **Keyword Search:** Finds problems using keywords related to problem names, categories, and symptoms.
- **Step-by-Step Solutions:** Provides clear instructions to help users troubleshoot selected problems.
- **Safety Reminders:** Displays warnings to help users avoid unsafe troubleshooting actions.
- **User Feedback:** Allows users to indicate whether a solution worked, worked partially, or did not work.
- **Input Validation:** Handles invalid menu selections, incorrect problem numbers, unavailable categories, and invalid feedback.
- **Exit Option:** Allows users to close the program safely.

5. Technologies Used

Python

Python is the main programming language used to develop the application. It handles the menu, search function, category browsing, troubleshooting instructions, and user feedback.

Jupyter Notebook

Jupyter Notebook provides an interactive environment for writing, running, and testing Python code.

IBM LinuxONE

IBM LinuxONE is the computing platform associated with this project. It provides an enterprise computing environment. The application can be run in a compatible Python environment on LinuxONE, subject to the available system configuration.

6. How to Run the Program

Requirements

- Python 3
- Jupyter Notebook or another Python-compatible environment
- Access to the project files

Steps to Run

1. Open Jupyter Notebook or your chosen Python environment.
2. Create a new Python notebook or open the notebook containing the project code.
3. Copy or load the IT Troubleshooting Assistant source code.
4. Run the code cell containing the program.
5. Wait for the welcome message and main menu to appear.
6. Enter a number from 1 to 4 to select an option.

Main Menu Options

Option	Description
1	Browse all problems
2	Browse by category
3	Search for a problem
4	Exit the program

Follow the instructions displayed on the screen to select a problem and view its troubleshooting guide.

Note: If the code is saved as a Python file named it_troubleshooting_assistant.py, it can also be run from a terminal using:

python it_troubleshooting_assistant.py

7. Team Members

The project team members are:

1. **DACHAINI A/P MURUGAIYIAH**
2. **LOSHENNI A/P MURUGIAH**
3. **DRIPTIKAA A/P BUVANESWARAN**
4. **THIRUSELVAN A/L BASKARAN**

8. Project Limitations

The current version has the following limitations:

- It contains only eight predefined troubleshooting problems.
- The search function uses basic keyword matching and may not recognise every description of a problem.
- Troubleshooting information is stored directly in the Python source code.
- The system cannot automatically detect or diagnose the actual condition of a computer.
- User feedback is displayed during the session but is not permanently stored.
- The application uses a text-based interface and does not include a graphical user interface.
- Complex technical problems may still require assistance from a qualified technician.

9. Future Improvements

Future versions could include a larger troubleshooting database, improved keyword matching, a graphical user interface, permanent feedback storage, multilingual support, and more advanced troubleshooting recommendations.

10. Conclusion

The IT Troubleshooting Assistant provides a simple way for students to find guidance for common computer problems. By offering searchable troubleshooting information, step-by-step solutions, and safety reminders, the system aims to reduce delays in resolving basic technical issues and help students become more independent when solving computer problems.